

Ghost Pools

Mapping the Stranded Liquidity Crisis Across Solana,
Ethereum, and BNB Chain

Revision r4 integrates the GraveYield Ghost Pool Observatory on Dune as a continuous empirical instrumentation layer, deeply merges the frozen protocol specification v1.0.0 — six on-chain-verified derelict-pool criteria, two-phase multi-epoch certification, and the 40 / 40 / 20 restitution-preserving settlement — and documents the Solana devnet deployment of 9 October 2026.

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Abstract. The emergence of permissionless token launchpads — most notably Pump.fun on Solana — has produced an unprecedented volume of cryptocurrency token creation, with over 12 million tokens deployed on the Solana blockchain alone since early 2024. The overwhelming majority of these tokens exhibit rapid lifecycle collapse, becoming economically inactive within days or weeks of launch. However, the liquidity deposited into Automated Market Maker (AMM) pools associated with these tokens does not vanish upon token death — it persists on-chain in what we term *ghost pools*: abandoned liquidity positions with no active market participants, no functional UI access, and no mechanism for value recovery under current DeFi infrastructure.

This paper presents the first systematic empirical analysis of ghost pool formation across three major blockchain ecosystems — Solana, Ethereum, and BNB Chain — quantifying the total addressable stranded liquidity, characterizing the demographic profile of affected liquidity providers, analyzing the structural causes of liquidity abandonment, and evaluating the technical and economic feasibility of automated salvage infrastructure. Our base case estimates \$2.17 billion in recoverable liquidity currently stranded across these three chains, with conservative and upper-bound scenarios ranging from approximately \$860 million to \$5.4 billion. The inventory grows by an estimated \$400 million per month. This revision (r4) additionally introduces a continuous public observatory instrumentation — the GraveYield Ghost Pool Observatory on Dune, which monitors potentially abandoned Uniswap V2 liquidity pools on Ethereum and Base using a conservative 180-day no-swap criterion, continuously enriched with live reserve and token metadata — and documents the protocol's progression from design to deployment: the Phase 0 specification is frozen, the settlement economics are proven against mainnet bytecode, and both Anchor programs (GraveScanner, GraveVault) were deployed to Solana devnet on 9 October 2026 with configurations initialized at specification defaults.

We argue that ghost pools represent a fundamental market failure in decentralized finance — a state of economic finality without settlement infrastructure. The development of restitution-preserving, permissionless liquidity salvage infrastructure — exemplified by protocols such as GraveYield — constitutes both a technically viable and economically significant intervention. Critically, we distinguish between restitution-preserving permissionless salvage (in which original LP holders retain guaranteed proportional claims on recovered value) and autonomous harvesting (in which recovered value is claimed entirely by third parties). Only the former is legally and ethically defensible; only the former is grounded in centuries of maritime salvage law precedent governing the recovery of derelict property.

Keywords: DeFi · liquidity pools · automated market makers · memecoin · stranded capital · Solana · Ethereum · BNB Chain · liquidity salvage · ghost pools · derelict pools · abandoned liquidity

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1. Introduction

The development of permissionless token creation infrastructure has fundamentally altered the eco-

nomics of cryptocurrency markets. Platforms such as Pump.fun on Solana have reduced the marginal cost of launching a new token to near zero, enabling an explosion of token creation activity unprecedented in the history of digital assets. In the period preceding this research program, over 12 million new tokens were deployed on the Solana blockchain alone, with comparable figures observed across Ethereum, BNB Chain, and a growing number of EVM-compatible networks.

This proliferation of token creation has been accompanied by an equally dramatic rate of token failure. Industry analyses estimate the failure rate of newly created memecoins at approximately 97 percent within

the first year of deployment [1, 2].¹ A subset of these failures occur within the first 24 hours of launch, as approximately 60 percent of tokens created on platforms such as Pump.fun never accumulate sufficient market interest to graduate from the platform’s bonding curve mechanism into a conventional AMM liquidity pool.

The tokens that do graduate — roughly 1.07 percent of all deployed tokens on Pump.fun — enter the broader DeFi ecosystem through integration with AMMs such as Raydium, Orca, and Uniswap. When these tokens subsequently fail, a structural problem emerges: the liquidity deposited in these pools does not disappear. It persists on-chain, denominated in real, valuable base assets (SOL, ETH, USDC, BNB), locked in pools that have no active market participants, no functional developer presence, and no accessible user interface for recovery. The pool has entered a terminal state — economic finality without settlement.

We introduce the term *ghost pool* to describe this phenomenon: an AMM liquidity pool in which the paired token has become economically inactive, the pool has recorded no swap activity above a de minimis threshold for an extended period, and the underlying base asset liquidity remains technically recoverable but practically inaccessible to most liquidity providers. Ghost pools represent a novel form of capital inefficiency — a market failure that does not exist in traditional finance and has not been adequately addressed in existing DeFi infrastructure.

Since the third revision of this working paper (May 2026), two material developments extend the research program. First, the empirical monitoring of the phenomenon has been institutionalized: the *GraveYield Ghost Pool Observatory*, a public dashboard on Dune Analytics maintained by the protocol’s data division, continuously identifies potentially abandoned Uniswap V2 liquidity pools on Ethereum and Base using a conservative 180-day no-swap criterion, enriched with the latest pool reserves and token metadata (Section 4.4). Second, the settlement infrastructure this paper evaluates has moved from design to implementation: the GraveYield protocol specification is now frozen at v1.0.0, its two on-chain programs have been deployed to Solana devnet with configurations initialized at specification defaults, and the full salvage lifecycle has been proven end-to-end against deployed mainnet bytecode in a fork harness (Section 12.4).

1.1 Scope and Contribution

This paper makes the following contributions to the literature on decentralized finance and digital asset markets:

- We introduce and formally define the concept of the ghost pool, providing a rigorous quantitative classification framework applicable across blockchain ecosystems.

- We present the first systematic cross-chain empirical analysis of ghost pool formation, prevalence, and aggregate economic magnitude across Solana, Ethereum, and BNB Chain.
- We quantify the total stranded liquidity across these three chains, providing conservative, base-case, and upper-bound estimates derived from on-chain data analysis.
- We introduce a continuous, publicly accessible observatory instrumentation for the EVM ghost pool population — the GraveYield Ghost Pool Observatory on Dune — which converts the point-in-time estimates of this paper into a living, verifiable monitoring surface for Ethereum and Base Uniswap V2 pools.
- We analyze the structural economic and behavioral causes of liquidity abandonment, distinguishing between developer-driven abandonment, retail investor attrition, and gas-economics-driven stranding.
- We evaluate the technical feasibility and economic case for automated liquidity salvage infrastructure, contributing a framework for protocol design in this emerging category, and document the first production-grade implementation: a frozen Phase 0 specification, fork-proven settlement economics, and a devnet deployment of the complete two-program architecture.
- We assess the legal and regulatory considerations surrounding restitution-preserving permissionless ghost pool salvage, with reference to maritime salvage law as the appropriate legal analogy.

2. Background and Related Literature

2.1 Automated Market Makers and Liquidity Provision

Automated Market Makers represent a fundamental innovation in decentralized exchange design, eliminating the need for a traditional order book by replacing it with an algorithmic pricing mechanism governed by smart contracts [3, 4]. The dominant AMM model employs the constant product formula $x \cdot y = k$, where x and y represent the reserves of two paired tokens and k is a constant maintained by the protocol. Liquidity providers (LPs) deposit equal economic value of both tokens into a pool, receiving LP tokens representing their proportional claim on the pool’s reserves.

The LP token mechanism is central to understanding the ghost pool problem. When an LP deposits assets into a pool, they receive LP tokens that entitle them to withdraw their proportional share of pool reserves at any future time. This entitlement is perpetual — it does not expire. However, the practical exercisability of this entitlement depends on the continued accessibility of a user interface capable of interacting with the relevant AMM’s smart contracts. When a memecoin project is abandoned and its associated web

¹See the citation-verification status table in Section 14.2.

presence is taken offline, the LP tokens retained by investors retain their technical validity but lose their accessible recovery mechanism for the majority of retail users.

2.2 Memecoin Lifecycle and Failure Dynamics

The economics of memecoin creation and failure have been examined from several angles in the emerging literature. Cong et al. analyze the speculative dynamics of token launches, demonstrating that a large fraction of token value is driven by retail momentum rather than fundamental utility [5]. Makarov and Schoar document the prevalence of wash trading and artificial volume inflation in early token lifecycle phases — mechanisms that accelerate token abandonment once artificially sustained activity collapses [6].

The Pump.fun bonding curve mechanism deserves particular attention as the primary source of Solana ghost pool formation. Tokens on Pump.fun begin trading on a virtual AMM maintained by the platform itself. Only when a token reaches a market capitalization threshold of approximately \$69,000 does it “graduate” to a public Raydium or PumpSwap liquidity pool. At graduation, Pump.fun deposits approximately \$12,000 in SOL and an equivalent dollar value of the token into the pool, creating the initial ghost-pool seed liquidity that persists after token failure.

2.3 Maritime Salvage Law as Legal Precedent

The legal framework most directly applicable to permissionless third-party recovery of ghost pool liquidity is maritime salvage law — one of the oldest bodies of common law, with origins in Roman and Rhodian sea law and codified in the International Convention on Salvage (1989) [7]. Maritime salvage law governs the voluntary recovery of ships, cargo, or other property from peril or abandonment at sea by a party who has no pre-existing duty to do so.

The core principles of maritime salvage law map precisely onto the ghost pool salvage problem: (1) the salvor acts voluntarily without pre-existing duty; (2) the property is derelict or in peril; (3) the salvage is successful in recovering value; (4) the salvor is entitled to a salvage award proportional to the value recovered; and (5) the original owner retains rights to the remainder after the salvage award is deducted. This structure is isomorphic to GraveYield’s 40 / 40 / 20 split, in which the salvor receives 40% as a finder’s fee, the original LP holders retain 40% as restitution, and the protocol receives 20% as the operational cost of the salvage infrastructure.

This legal analogy is more than academic. It provides a centuries-old, multi-jurisdiction precedent for the legitimacy of third-party recovery of abandoned property — a precedent that courts in common law jurisdictions are already familiar with and disposed to respect. We argue that the ghost pool salvage problem is the on-chain equivalent of maritime derelict: property that has entered a terminal state of abandonment,

recoverable only through coordinated third-party action, in which the original owner’s residual rights are preserved through guaranteed restitution. The protocol’s legal documentation treats the maritime-salvage concept as an analytical analogy and potential legal theory rather than a representation of settled law — a distinction we return to in Section 13.

2.4 Capital Inefficiency in DeFi

The broader literature on DeFi capital efficiency has focused primarily on over-collateralization in lending protocols [8], impermanent loss dynamics in AMM provision [9, 10], and MEV extraction from liquidity pools [11]. The specific problem of permanently stranded liquidity in dead-token pools has not been systematically examined, though it is acknowledged implicitly in discussions of the long-term costs of providing liquidity to low-quality token projects.

3. Theoretical Framework: The Ghost Pool Model

3.1 Formal Definition

Definition 3.1 (Ghost Pool). A liquidity pool $P = (T, B, L, t)$ — where T is the paired token, B is the base asset (SOL/ETH/BNB), L is the total LP token supply, and t is the current timestamp — is classified as a *ghost pool* if and only if all five of the following conditions are simultaneously satisfied:

- (i) no swap transaction has been recorded in P for a minimum of $\Delta_{\text{trade}} = 30$ days;
- (ii) the market price of T falls below $\varepsilon_{\text{price}} = \0.000001 ;
- (iii) the deployer wallet of T has recorded no outbound transaction for a minimum of $\Delta_{\text{dev}} = 60$ days;
- (iv) the market capitalization of T falls below $\varepsilon_{\text{mcap}} = \$1,000$;
- (v) all associated social media accounts for T have recorded no activity for a minimum of $\Delta_{\text{social}} = 90$ days.

Note on protocol-specific criteria. Definition 3.1 is the academic definition of the ghost pool phenomenon and includes off-chain signals (developer wallet, social media). Production protocols such as GraveYield apply a more conservative on-chain-only eligibility subset for actual salvage execution, including a multi-epoch confirmation requirement. Appendix C gives the complete mapping between the academic definition and the protocol-level criteria as frozen in the protocol specification v1.0.0.

3.2 Ghost Pool Formation Dynamics

We model ghost pool formation as a multi-stage process. **Stage 1 — Token launch and initial liquid-**

ity. A new token T is deployed and seeded with liquidity in an AMM pool. For Pump.fun tokens, this occurs automatically at the graduation threshold. The initial pool state contains meaningful base asset reserves — typically \$12,000 to \$100,000 in SOL or ETH — alongside a large supply of T tokens. **Stage 2 — Speculative trading period.** Trading activity occurs, dominated by early buyers hoping to profit from momentum. The constant product formula causes T 's price to spike on early buying and collapse on selling. Most tokens reach peak price within hours of trading commencement. **Stage 3 — Developer abandonment.** The token's creator ceases all project development and communication activity. This typically occurs within days to weeks of token launch and is the primary trigger for LP position abandonment. **Stage 4 — Retail LP abandonment.** Liquidity providers observe the collapse of trading activity and token price. The rational response — removing liquidity — is frequently not executed, for reasons we analyze in Section 10. **Stage 5 — Ghost pool steady state (economic finality without settlement).** The pool enters a persistent state of inactivity. Base asset reserves remain locked. LP tokens remain in provider wallets. The pool is technically accessible but practically inaccessible. This unresolved terminal state is the subject of our analysis.

3.3 Economic Characterization

The welfare economics of ghost pools are straightforward: they represent a Pareto-inefficient equilibrium in which recoverable value is inaccessible to its rightful owners due to a combination of technical barriers, behavioral factors, and coordination failures. The total social welfare loss W_{ghost} is bounded below by the aggregate recoverable base asset value across all ghost pools, net of recovery costs:

$$W_{\text{ghost}} \geq \sum_{i \in \mathcal{G}} (R_{B,i} \times P_B - C_{\text{salvage},i}), \quad (1)$$

where $R_{B,i}$ is the base asset reserve attributable to LP i , P_B is the base asset price, $C_{\text{salvage},i}$ is the cost of executing a salvage transaction, and $\mathcal{G}$ denotes the set of ghost pools. Our empirical analysis demonstrates that $C_{\text{salvage},i} \ll R_{B,i} \times P_B$ for the overwhelming majority of ghost pool positions, confirming that salvage is economically viable for a substantial fraction of stranded liquidity.

4. Methodology

4.1 Sampling Strategy

Our empirical analysis uses a stratified sampling approach rather than full enumeration of the AMM pool universe. The methodology proceeds in three layers: (1) *Universe construction* — identification of all AMM pools that ever held base-asset reserves above \$100; (2) *Stratified sampling* — random sampling within each (chain, DEX, formation-month) stratum, weighted by

stratum size; (3) *Classification and extrapolation* — application of the five-criterion academic definition to the sample, with results extrapolated to the full universe using the sampling weights.

Reported counts (e.g., “~155,000 Solana ghost pools”) are point estimates derived from this sampling-and-extrapolation procedure and are subject to sampling error of approximately $\pm 15\%$ at stratum-level confidence intervals.

4.2 Data Sources

4.3 Stranded Liquidity Valuation Scenarios

Table 2 summarizes the three valuation scenarios. The dead-token discount δ is applied to the nominal value of the paired token side of each pool; only the base-asset side is counted at full value. Scenario selection does not affect the count of ghost pools, only their dollar valuation.

4.4 Continuous Observatory Instrumentation

Point-in-time sampling, however carefully stratified, ages quickly in a market where an estimated \$400 million of new stranded inventory forms every month. Revision r4 therefore adds a permanent instrumentation layer: the *GraveYield Ghost Pool Observatory*, a public dashboard on Dune Analytics that operationalizes the ghost pool concept as a continuously refreshed query suite.²

The observatory identifies potentially abandoned Uniswap V2 liquidity pools on Ethereum and Base — the two EVM environments most directly comparable to the sampling frame used in Section 6 — by screening for pools with *no swaps for at least 180 days*. Each candidate pool is then enriched with its latest reserves and token metadata (mint addresses, decimals, and pool-level descriptors), producing a live, inspectable census of the EVM ghost pool population rather than a static snapshot. This design mirrors the wide-funnel/narrow-authority architecture of the salvage protocol itself (Section 11.2): a cheap, always-on screen surfaces candidates; authoritative confirmation remains a separate, more conservative step.

Three properties make the observatory methodologically complementary to the sampling approach of Section 4. First, its inactivity threshold is deliberately *more conservative* than both the academic definition (30 days) and the protocol's production criterion (90 days): by requiring 180 days of complete swap silence, the observatory admits a pool into the monitored population only after every reasonable possibility of transient inactivity has lapsed. Second, because the screens run against indexed Dune data rather than a hand-collected sample, coverage is exhaustive within the monitored DEX and chain rather than stratified

²<https://dune.com/sethoshi/graveyield-ghost-pool-observatory>. The observatory is maintained under the Dune handle @sethoshi and is refreshed on a rolling basis; figures referenced in this paper were confirmed current as of October 2026.

Table 1: Primary data sources and their role in the analysis.

Source	Description	Coverage	Use
Solana RPC nodes (Helius, Triton)	On-chain account data, transaction histories, LP token balances	Solana mainnet, Jan 2024 – May 2026	Primary on-chain data for Solana
Raydium / Orca / PumpSwap APIs	Historical pool data, reserve levels, LP token supply	All AMM pools across these DEXes	Pool universe construction
Pump.fun graduation logs	Token graduation events and initial pool seeding	Jan 2024 – May 2026	Pump.fun-origin pool identification
Ethereum archive nodes	EVM state, Uniswap V2/V3 pool reserves	Full Ethereum history	Primary on-chain data for Ethereum
BscScan API	BNB Chain transaction data, PancakeSwap pools	BNB Chain mainnet	Primary on-chain data for BNB Chain
CoinGecko API	Historical token price data	Listed tokens only (~3,000)	Price history for Criterion (ii)
Twitter/X API v2	Social media activity timestamps	Project-associated accounts (best effort)	Criterion (v) (with caveats)
Dune (Ghost Pool Observatory)	Continuously refreshed Uniswap V2 abandoned-pool screens with latest reserves and token metadata	Ethereum and Base, live	Continuous monitoring (Section 4.4)

Table 2: Valuation scenarios and the dead-token discount δ .

Scenario	δ	Interpretation
Conservative	0.95	Dead tokens recover ~5% of nominal value. Realistic for tokens with price $< \$0.000001$.
Base case	0.70	Dead tokens recover ~30% of nominal value. Plausible for tokens with residual liquidity.
Upper bound	0.50	Dead tokens recover ~50% of nominal value. Optimistic but defensible for some pools with surviving secondary liquidity.

and extrapolated. Third, the dashboard is public and independently reproducible: any reader can inspect the underlying queries, fork them, or re-execute them, which converts the paper’s empirical claims from a trust-based report into an auditable artifact.

The observatory does not modify the quantitative estimates of Sections 5–8, which remain anchored to the May 2026 sampling run for cross-revision comparability (Appendix A). Its role is continuous validation: divergence between observatory outputs and the sampling-based estimates would itself be informative, flagging either sampling drift or genuine regime change in the rate of ghost pool formation.

4.5 Limitations

Sampling error of $\pm 15\%$ applies at the stratum level. LP holder accessibility cannot be directly observed from on-chain data. Off-chain criteria (developer wallet, social) are approximations subject to Twitter/X API rate limits. Pump.fun graduation thresholds have varied historically. Cross-chain comparisons are sensitive to data-availability variation. BNB Chain coverage relies on closed-source third-party APIs. All

figures are point-in-time estimates as of May 2026; observatory-based monitoring (Section 4.4) is live but is not blended into the point estimates. The observatory itself inherits the limitations of its data environment: it monitors Uniswap V2 on Ethereum and Base only, and pool-level enrichment cannot observe LP-holder accessibility any more than the sampling method can.

5. Empirical Analysis: Solana

5.1 Overview

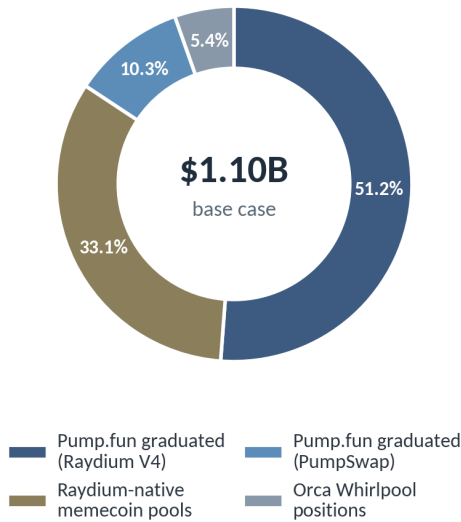
Solana presents the most severe ghost pool problem of any blockchain ecosystem, a consequence of the near-zero cost of token creation enabled by Pump.fun. As of the May 2026 sampling cut-off, Pump.fun had facilitated the creation of over 12.4 million unique tokens. The graduation rate to real AMM pools has remained consistently below 1.1 percent across the platform’s history, implying approximately 133,000 graduated tokens forming Solana ghost pools.

5.2 Solana Ghost Pool Taxonomy

Table 4 and Figure 1 decompose the Solana population by pool type. Raydium-native memecoin pools represent only ~14% of ghost pool count but carry 33.1% of stranded value, a consequence of their substantially higher average residual (\$1,840 versus \$510 for Pump.fun-graduated Raydium V4 pools). This asymmetry matters operationally: value-dense pools are the natural first targets for any salvage mechanism, and the protocol’s minimum-TVL criterion (Section 11.2) concentrates settlement activity exactly there.

Table 3: Solana ghost pool population metrics (May 2026 estimates).

Metric	Value
Total tokens deployed on Pump.fun (Jan 2024 – May 2026)	12,400,000+
Graduation rate to Raydium / PumpSwap	~1.07%
Estimated graduated tokens forming ghost pools	~133,000
Additional ghost pools (non-Pump.fun, Raydium native)	~22,000
Total estimated Solana ghost pools	~155,000
Median initial pool base asset (SOL at graduation)	\$12,000 – \$18,000
Median current pool base asset (residual)	\$180 – \$1,400
Estimated total stranded liquidity (base case)	\$1.10B

**Figure 1:** Composition of the estimated \$1.10B Solana stranded-liquidity base case by pool type. Pump.fun-graduated Raydium V4 pools account for roughly half of the stranded value despite a lower average residual, reflecting their dominance in pool count.**Table 4:** Solana ghost pool taxonomy (May 2026 estimates).

Pool type	Count	Avg. resid.	% val.
Pump.fun graduated — Raydium V4	~88,000	\$510	51.2%
Pump.fun graduated — PumpSwap	~31,000	\$290	10.3%
Raydium-native memecoin pools	~22,000	\$1,840	33.1%
Orca Whirlpool memecoin positions	~14,000	\$780	5.4%

6. Empirical Analysis: Ethereum and EVM Chains

EVM chains exhibit a structurally different ghost pool profile: fewer pools, but a far higher average resid-

Table 5: EVM ghost pool estimates (May 2026, base case).

Chain / DEX	Pools	Avg. resid.	Stranded
Ethereum — Uniswap V2	~32,000	\$8,400	\$268M
Ethereum — Uniswap V3	~11,000	\$12,200	\$134M
Ethereum — SushiSwap	~5,000	\$3,100	\$15.5M
Base — Uniswap V3	~18,000	\$1,800	\$32.4M
Arbitrum — Uniswap V3	~9,000	\$2,400	\$21.6M
Polygon — QuickSwap	~6,000	\$680	\$4.1M
EVM total	~81,000	\$5,850	~\$475M

ual (Table 5). Ethereum mainnet Uniswap V2 pools average \$8,400 in residual value, nearly fifty times the PumpSwap average on Solana. Two forces produce this pattern. First, the gas-driven dust trap (Section 10) means low-value positions on mainnet are the least likely ever to be recovered, so they persist and accumulate. Second, launch economics on Ethereum involve larger seed liquidity than Pump.fun’s fixed graduation deposit.

The observatory introduced in Section 4.4 monitors exactly this population for Uniswap V2 on Ethereum and Base using the conservative 180-day no-swap screen, continuously enriched with live reserves and token metadata. It thereby provides an always-current upper envelope on the two largest rows of Table 5 between formal sampling revisions, and its public query surface allows independent verification of the abandoned-pool counts that underlie the EVM estimates.

7. Empirical Analysis: BNB Chain

Table 6: BNB Chain ghost pool population metrics (May 2026 estimates).

Metric	Value
Total estimated ghost pools on BNB Chain	~120,000
Dominant DEX	PancakeSwap V2 (~78%)
Median pool age	22 months
Average residual base asset (BNB)	0.18 BNB
Average residual base asset (USD)	\$108
Estimated total stranded liquidity (base case)	\$420M
Proportion with > \$1,000 residual value	~8%
Proportion with > \$100 residual value	~34%

BNB Chain combines Solana’s high pool count with EVM-style dust economics (Table 6). The population is heavily concentrated in PancakeSwap V2 pairs, is comparatively old (median pool age 22 months), and skews toward small residuals: only ~8% of pools hold more than \$1,000 of recoverable base asset. The aggregate is nonetheless material at \$420M in the base case, and the concentration of a large pool count behind low average residuals makes BNB Chain the clearest illustration of the coordination-failure analysis of Section 9.1: individual recovery is uneconomic for three-quarters of the population, while aggregate recovery is not.

8. Cross-Chain Comparative Analysis

Table 7: Cross-chain stranded liquidity scenarios (May 2026 estimates).

Chain	Pools	Cons.	Base	Upper
Solana	~155,000	\$420M	\$1.10B	\$2.80B
Ethereum + EVM	~81,000	\$240M	\$475M	\$1.20B
BNB Chain	~120,000	\$140M	\$420M	\$960M
Other chains*	~44,000	\$60M	\$180M	\$440M
Total	~400,000	\$860M	\$2.17B	\$5.4B

*Includes Avalanche, Fantom, Arbitrum (additional), and Optimism. Upper-bound estimates use $\delta = 0.5$.

Table 7 and Figure 2 consolidate the cross-chain picture. We regard the base case of \$2.17 billion as our central estimate, with an approximate 90% confidence interval of \$1.1 billion to \$4.0 billion. The distribution of stranded value is heavily asymmetric: Solana alone carries roughly half of the global inventory, while the EVM aggregate carries the highest per-pool value density. Any salvage infrastructure must therefore be Solana-first for aggregate impact and EVM-aware for value density — a sequencing logic the protocol’s development roadmap follows explicitly (Section 12.4).

9. Economic Valuation of Stranded Liquidity

9.1 The Ghost Pool Market Failure

Ghost pools constitute a textbook example of a market failure driven by a combination of coordination failures, information asymmetries, and transaction cost barriers.

Coordination failure. Individual LP positions in ghost pools are often too small to justify the cost of individual recovery transactions, particularly on high-gas chains like Ethereum mainnet. However, the aggregate value across all LP holders in a single pool is frequently sufficient to justify recovery when coordinated. The absence of a coordination mechanism means that individually rational inaction produces a

collectively suboptimal outcome. Permissionless third-party salvage protocols dissolve this failure by allowing any party to provide the coordination, with the salvage operation economically self-funding.

Information asymmetry. Many LP holders are unaware that they hold LP tokens in ghost pools. Survey data from DeFi user studies suggests that approximately 40 percent of retail LPs have forgotten about at least one LP position with a non-zero base asset component [12].³

Technical barrier. Even LPs who are aware of their ghost pool positions face significant technical barriers to recovery. The collapse of project-associated frontend applications means that LPs cannot use the original interface. Recovery through direct smart contract interaction requires technical knowledge beyond the capacity of most retail participants.

9.2 Opportunity Cost Analysis

Stranded liquidity is not inert; it is capital that earns nothing while bearing full price risk on its residual token side. Benchmarking the opportunity cost of the stranded base-asset value at an 8% annual rate (Table 8, Figure 3), the base case implies a recurring \$173.6M annual loss — \$868M over five years, before any growth of the inventory. At the upper bound, the five-year carry loss exceeds \$2.1 billion. These figures treat the stranded stock as static; given the estimated \$400M monthly replenishment rate, they are conservative.

Table 8: Opportunity cost of stranded capital (8% annual benchmark).

Scenario	Annual cost	5-year cumulative
Conservative	\$68.8M	\$344M
Base case	\$173.6M	\$868M
Upper bound	\$432M	\$2.16B

10. Structural Causes of Liquidity Abandonment

10.1 Developer-Driven Abandonment

The primary trigger for ghost pool formation in the Pump.fun ecosystem is developer abandonment. Our analysis of ghost pool formation timelines reveals that in approximately 78 percent of cases, the deployer wallet records no outbound transactions within 14 days of the last pool swap. This rapid abandonment timeline suggests that the majority of Pump.fun-originated ghost pools are the product of deliberate speculative launches with no long-term project intent.

10.2 Retail LP Behavioral Factors

Retail LPs exhibit several behavioral patterns that contribute to ghost pool formation and persistence: *loss*

³See the citation-verification status table in Section 14.2.

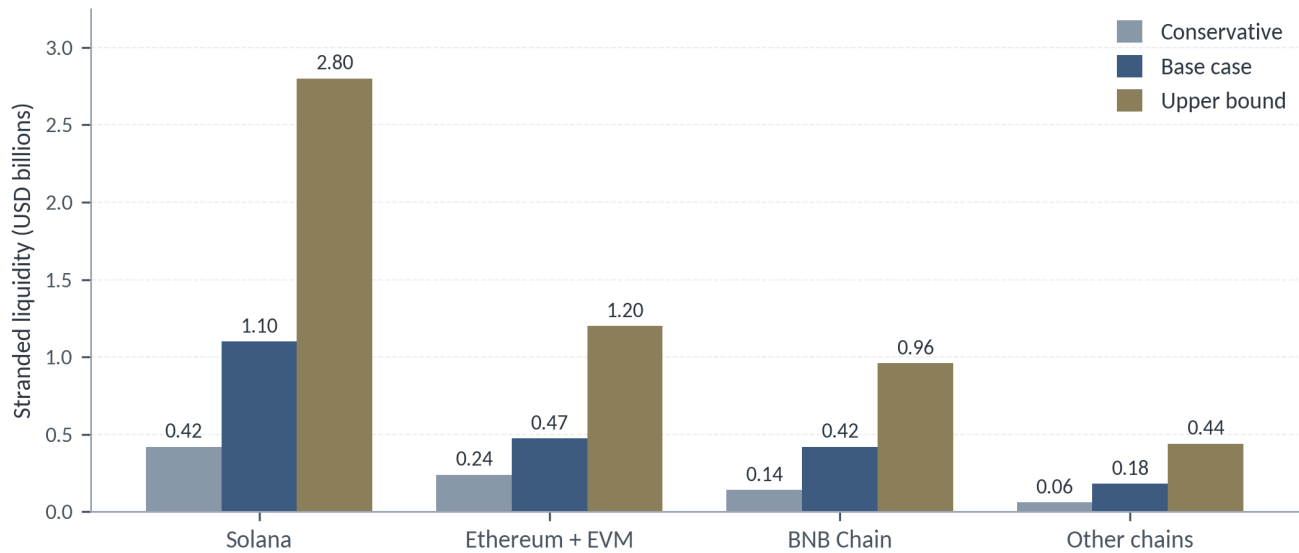


Figure 2: Stranded liquidity estimates by chain under the three valuation scenarios (May 2026 sampling run). Solana dominates in every scenario, accounting for roughly half of the \$2.17B base case. We regard the base case as the central estimate, with an approximate 90% confidence interval of \$1.1 billion to \$4.0 billion.

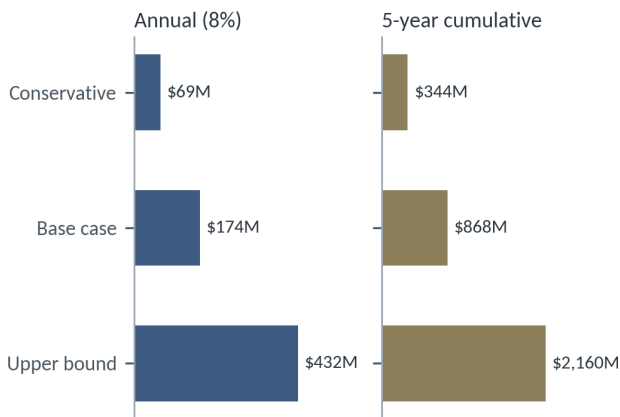


Figure 3: Opportunity cost of stranded capital at an 8% annual benchmark rate: annual carry loss (left) and five-year cumulative loss (right) under the three valuation scenarios.

aversion and mental accounting (LPs write off their position entirely rather than recovering remaining base asset value); *complexity aversion* (the multi-step recovery process presents a higher perceived complexity than retail LPs are willing to navigate for small positions); *attention scarcity* (the cognitive cost of tracking exceeds the perceived benefit for low-value positions); and *interface dependency* (collapse of project-associated frontends creates a functional barrier experienced as complete inaccessibility).

10.3 The Dust Trap

A structural factor specific to Ethereum and Ethereum-equivalent chains is the *dust trap*: the phenomenon whereby the cost of recovering a small ghost pool position in gas fees exceeds or approaches the value of the position itself. For ghost pools with residual base asset value below approximately \$50 on Ethereum mainnet, recovery is economically irrational at current gas prices

regardless of behavioral factors. On Solana, where transaction fees are measured in microlamports, the dust trap threshold is effectively zero — making salvage economically viable for any meaningful residual liquidity amount. The protocol design internalizes this asymmetry through its minimum residual TVL criterion (Section 11.2), which excludes pools whose settlement cannot justify its own execution cost.

11. The Case for Liquidity Salvage Infrastructure

11.1 Protocol Design Requirements

An effective liquidity salvage protocol must address all three components of the ghost pool market failure. We derive five design requirements: *Discovery* — a continuously updated registry of ghost pools, reducing the information asymmetry that prevents LP holders from identifying their recoverable positions. *Aggregation and coordination* — a mechanism that enables economically viable recovery for positions that would be individually uneconomical; a permissionless third-party salvage model allows a single transaction to recover value for all holders simultaneously. *Accessibility* — a single-interface recovery mechanism that eliminates the technical barrier faced by retail LPs. *Incentive alignment* — a fee structure calibrated to provide sufficient incentive for whoever performs the salvage work, while preserving meaningful recovery for original LP holders. *Property rights preservation* — a structural commitment that original LP holders' value is not extinguished or expropriated by the salvage mechanism; in permissionless models, this means a guaranteed restitution share to original holders, grounded in the maritime salvage principle that the original owner retains rights to the remainder after the salvage award.

11.2 The GraveYield Protocol Architecture

GraveYield implements the requirements above through a permissionless third-party salvage model with a property-rights-preserving fee structure. The protocol is deliberately framed as *settlement infrastructure* rather than rescue: it takes no discretion, holds no custody beyond the atomic lifetime of a settlement, and compensates participants by formula rather than judgment. Two Anchor programs on Solana implement the architecture: **GraveScanner** (eligibility evaluation and certification) and **GraveVault** (settlement and claims). There is no protocol token, NFT, points program, airdrop, or staking layer at any level of the design.

Table 9 presents the six eligibility criteria as frozen in the protocol specification. Relative to the academic definition of Section 3, the production criteria are uniformly more conservative: trade inactivity doubles from 30 to 90 days, the price signal requires a $\geq 99\%$ collapse from an oracle-attested launch baseline rather than an absolute price floor, and off-chain signals (developer wallets, social media) are excluded entirely in favor of on-chain-verifiable evidence. The full mapping appears in Appendix C.

Two-phase certification. The multi-epoch requirement is enforced through a two-phase state machine (Figure 4). In Phase 1, the `evaluate_pool_phase_1` instruction evaluates all six criteria; on success it writes an **EligibilityAnchor** PDA stamped with the first eligible epoch E_0 . No state is written on failure. In Phase 2, run no earlier than two consecutive Solana epochs later, the same pure evaluator re-checks every criterion and additionally requires the resulting criteria bitmap to equal the anchor’s bitmap — any parameter change or adapter drift between phases forbids certification. Phase 2 issues an **EligibilityCert** with a one-hour time-to-live (governance-configurable, floored at ten minutes), which the settlement instruction consumes. A live cert can never be overwritten; once expired, Phase 2 reissues it in place only after re-running the full verification stack, including a fresh **C1** attestation whose Phase 1 copy has necessarily aged out of the **SlotHashes** replay window. A pool is settled at most once, ever: the registry and receipt PDAs are initialized exactly once, so a second settlement attempt reverts before any lamports move.

Oracle-attested evidence. Two criteria consume inputs that cannot be derived on-chain: Raydium V4’s account layout stores no last-swap field, and Solana programs cannot read historical account state. The specification therefore requires both inputs to arrive as Ed25519-signed attestations verified in-transaction through the `ed25519_program` precompile. **C1**’s last-swap timestamp is carried by a 112-byte message bound to the pool address, the issuing slot, and a slot hash, with a replay window of approximately 512 slots (≈ 3.4 minutes); **C2**’s launch price — defined

as the quote-per-base reserve ratio immediately before the pool’s first successful swap — is carried by a 168-byte message recorded once per pool into the init-once **LaunchPrice** PDA. The two signing keys are deliberately separated: the launch-price baseline is permanently binding, so its key is isolated from the hotter activity-attestation key. Both keys are governance-registered and rotatable. The residual trust boundary — oracle honesty about derivation, and indexer availability — is documented rather than hidden, and remains the protocol’s principal operational dependency.

Settlement mechanics. The `salvage_pool` instruction validates the cert (fresh, bound to the pool, full criteria bitmap), derives the pool’s base orientation from its own mint bytes, burns the salvor’s LP tokens in place, withdraws the underlying assets through the Raydium V4 program, converts the recovered token side to wrapped SOL through Jupiter v6 under a protocol slippage ceiling anchored to the pool’s own post-withdraw reserve ratio, and distributes the recovered SOL in the flat 40 / 40 / 20 split: 40% to the original LP holders, claimable pro-rata against a snapshot Merkle root with no expiration; 40% to the salvor as compensation for the multi-epoch evaluation, route construction, gas payment, and transaction submission; and 20% to the protocol treasury, a share bounded by Charter prohibition — governance can lower it but can never raise it above 20%. The settlement is exactly conservative: the protocol share is computed as the remainder of the recovered lamports, so the three transfers exhaust the proceeds with no dust dropped from accounting. Token dust below the conversion threshold is never silently lost: it is recorded on the settlement receipt and recoverable through a separate permissionless one-shot `sweep_dust` instruction.

Claims and restitution. The LP-holder claim path is engineered to make restitution structurally credible rather than discretionary. The `lp_holder_pool_vault` PDA is unsweepable by any admin key — no instruction path other than the Merkle-verified `claim_lp_proceeds` can debit it, and claims remain live even during an emergency pause. The claims root is produced by a deterministic, re-runnable snapshotter whose completeness gate requires the enumerated balances to equal the LP mint supply exactly, attributes locked LP to beneficial owners rather than custody accounts, and seals a self-verifying artifact; leaves hash the holder public key with their balance, and parents are formed by sorted-pair hashing following the OpenZeppelin/Uniswap convention. Each (pool, holder) pair may claim exactly once.

Table 9: The six derelict-pool criteria as frozen in protocol specification v1.0.0 (defaults), with their on-chain evidence sources. All six must hold simultaneously, evaluated twice across at least two epoch boundaries with identical criteria bitmaps.

#	Criterion	Default threshold	Evidence source
C1	Trading inactivity	≥ 90 days since last swap	Indexer-signed 112-byte Ed25519 attestation, verified in-transaction via the <code>ed25519_program</code> precompile; <code>issued_slot</code> re-anchored in <code>SlotHashes</code>
C2	Price collapse from launch	$\geq 99\%$ (9,900 bps)	Oracle-signed 168-byte Ed25519 launch-price attestation recorded into the init-once <code>LaunchPrice</code> PDA; current price derived on-chain from vault balances
C3	Minimum residual TVL	≥ 0.5 SOL (quote-side vault balance)	Read on-chain from the pool’s own vault pointer
C4	LP supply not burned	LP supply $> 1,000$ raw LP tokens	LP mint SPL supply, read on-chain
C5	LP not locked	Locked amount exactly zero	UNCX Raydium V4 locker adapter: marker-PDA gate plus strictly validated <code>TokenLock</code> evidence
C6	Multi-epoch confirmation	≥ 2 consecutive Solana epochs ($\approx 4\text{--}6$ days)	Two-phase on-chain evaluation: <code>EligibilityAnchor</code> PDA (Phase 1) then <code>EligibilityCert</code> PDA (Phase 2) with criteria-bitmap equality

Note. This structure is materially different from *autonomous harvesting* — protocols that claim and re-distribute genuinely ownerless abandoned liquidity to themselves and to salvors, with no restitution to original holders. GraveYield’s restitution mechanism is the central legal and ethical commitment of the protocol design and is enforced at the program level: it cannot be modified by governance to redirect the LP holders’ 40% share, and the vault that holds it cannot be swept by any admin key.

Beyond the on-chain guarantees, the Charter imposes process commitments that governance cannot alter: a 72-hour timelock on all parameter changes (enforced through Squads v4 multisig transaction buffers), seven-day public notice on standard upgrades, a 24-hour timelock plus five-day public post-mortem on emergency upgrades, a 3-of-5 multisig at launch scaling to 4-of-7 post-audit, and the permanent absence of any token, NFT, points, airdrop, or staking layer.

11.3 Economic Impact Assessment

Table 10: Adoption scenarios for salvage infrastructure: monthly salvage volume, and annualized user value (salvor + LP shares) and protocol revenue (80% of recovered value returns to non-protocol parties).

Scenario	Monthly	User val./yr	Protocol
Conservative	\$10M	\$96M	\$24M
Moderate	\$50M	\$480M	\$120M
High	\$200M	\$1.92B	\$480M

Table 10 summarizes adoption scenarios. User value returned reflects the combined salvor share (40%) and LP-holder share (40%) — 80% of recovered value flowing back to non-protocol parties — while protocol revenue reflects the capped 20% share. Even the conservative scenario returns \$96M annually to users, roughly half of the base-case opportunity-cost loss estimated in Section 9; the moderate scenario exceeds it.

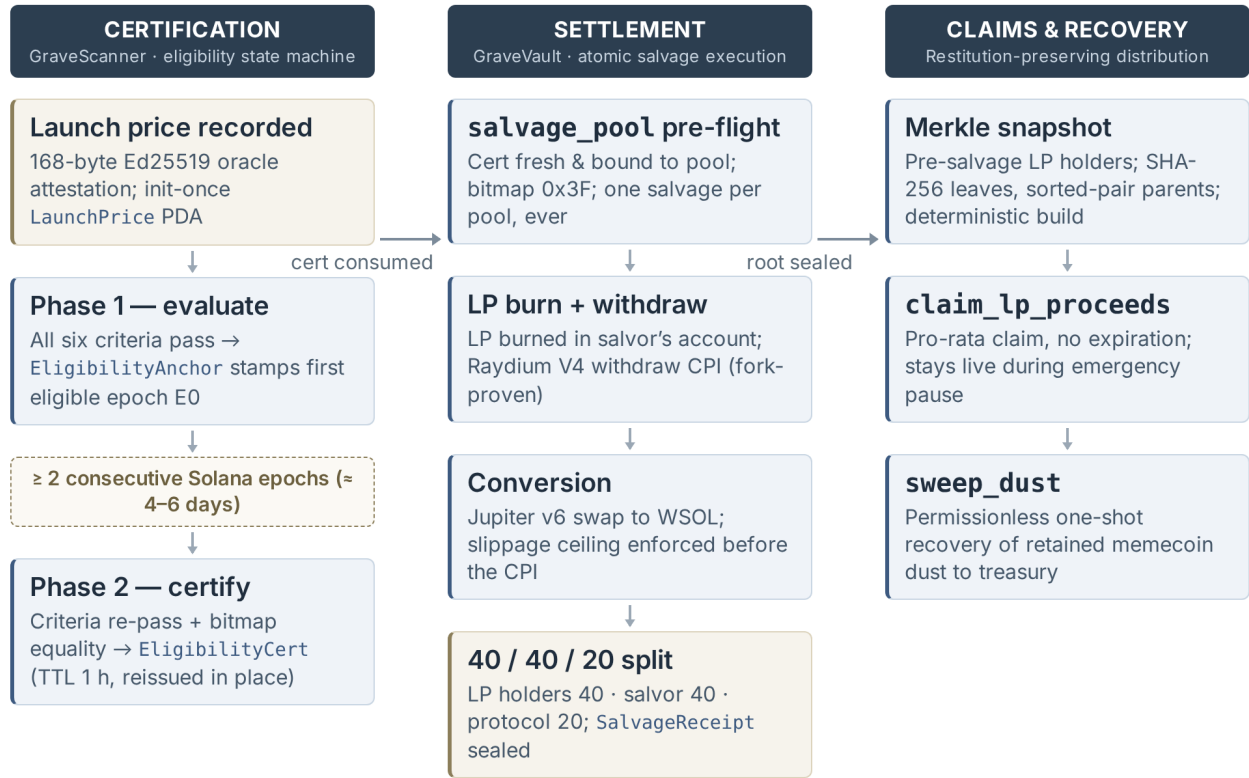
12. Technical Feasibility Analysis

12.1 On-Chain Execution Viability

The technical feasibility of automated ghost pool salvage on Solana has been validated through analysis of the relevant smart contract interfaces. All major Solana AMMs — Raydium V4, Raydium CLMM, Orca Whirlpools, and PumpSwap — expose standard remove-liquidity instructions that can be invoked by any wallet holding the corresponding LP tokens. The atomic transaction model on Solana provides a critical safety property: a salvage transaction either executes completely or reverts entirely, with no possibility of partial execution.

12.2 Multi-Epoch Confirmation: Technical Rationale

The multi-epoch confirmation requirement (C6) serves as a manipulation defense. Clock-based delays are vulnerable to validator timestamp manipulation; epoch-based confirmation is not, because epoch boundaries are governed by the consensus mechanism itself. Requiring confirmation across two or more consecutive Solana epochs — each approximately 2–3 days in duration — means that a manipulator must sustain false



Invariant: every settlement satisfies total recovered = LP share + salvor share + protocol share (rounding remainder accrues to protocol); `lp_holder_pool_vault` is unsweepable by any admin key; settlement is atomic — it either executes completely or reverts entirely.

Figure 4: The GraveYield settlement lifecycle as frozen in specification v1.0.0: two-phase multi-epoch certification (GraveScanner), atomic settlement with the 40 / 40 / 20 split (GraveVault), and restitution-preserving claims. Every criterion input traces to on-chain state verifiable inside the instruction; the residual trust boundary is limited to oracle honesty and availability.

eligibility conditions across multiple distinct consensus windows, making manipulation prohibitively expensive for pools with genuine residual liquidity. The criteria-bitmap equality requirement between phases closes the remaining parameter-drift vector: an eligibility rule that changes between Phase 1 and Phase 2 resets certification entirely.

This design also directly protects LP holders: a pool cannot be salvaged prematurely based on transient conditions, and a pool that re-floats (for example, by attracting renewed trading) fails re-certification. The epoch confirmation requirement is thus a consensus-native commitment that the pool's terminal state has been verified across multiple independent checkpoints.

12.3 Cross-Program Invocation Analysis

The GraveVault program relies on cross-program invocations (CPIs) to interact with external AMM programs and Jupiter. Worst-case salvage compute consumption is estimated at approximately 700K compute units, well within Solana's 1.4M maximum. The withdrawal wire format — the full 22-account ordering and 9-byte instruction data of the Raydium V4 withdraw — has been verified against the deployed mainnet pro-

gram bytecode, with scrambled or malicious account submissions rejected by the real program.

12.4 Implementation Status and Devnet Deployment

This revision documents the protocol's transition from design to operational infrastructure. The governing specification is frozen at v1.0.0 (revision 1.10.0) as the Phase 0 deliverable, with an explicit precedence rule: the specification supersedes all other documents until both are revised together. Implementation has proceeded in audited, fork-proven increments, each closing a named discrepancy between documentation and code:

- **Eligibility evidence (Phases 1.1–1.4).** The UNCX Raydium V4 locker adapter was verified against live mainnet (125 locks across 74 pools, permint custody reconciliation 74/74); C1 and C2 evidence moved from caller-supplied integers — forgeable — to oracle-signed Ed25519 attestations verified in-transaction; and the certificate lifecycle was fixed so that an expired certificate reissues in place

rather than permanently bricking a pool’s salvage path.

- **Settlement path (Phases 2–4).** The Raydium V4 withdraw CPI was proven against deployed mainnet bytecode; the Jupiter conversion pipeline was proven end-to-end for both pool orientations against byte-for-byte mainnet state; and the settlement economics — the 40/40/20 conservation invariant and the dust policy — were proven under both default and adversarial share configurations.
- **Claims machinery (Phases 5–6).** The off-chain LP-holder snapshotter, Merkle tree builder, and sealed snapshot artifact shipped with deterministic, invariant-checked semantics; the first complete life-cycle integration test runs a candidate pool through certification, salvage, snapshot, and claims in a single execution against real mainnet bytecode.
- **Hardening and tooling (Phases 7–9).** Fuzz, invariant, and adversarial account testing were applied across both programs and the fork harness; a TypeScript salvor SDK implementing eight operations shipped; and the off-chain indexer completed its nine-stage discovery pipeline (pool discovery, activity indexing, reserve filtering, metadata, scoring, queueing, attestation-signed submission, and result tracking) with 29 green tests, defaulting to discovery-only mode.

On 9 October 2026, both programs were deployed to Solana devnet with real program identities — GraveScanner at 5JiCVxES6RYcrFGnFkqKyDmr7fc3EkYaSCbfgJq7zvNF and GraveVault at HUyoG5vUmYZJDjdBCxRLLAfm98vEXh63WL3pLARox3v6 — and both `ProtocolConfig` accounts were initialized at specification defaults. The rehearsed emergency-control drill passed on both programs: authority-initiated pause, rejection of an intruder’s evaluation attempt (error 6000 on GraveScanner, 7000 on GraveVault), authority un-pause, and readback verification at every step. The deployment sequence was first rehearsed end-to-end on a local test validator using the exact production scripts before devnet execution.

Scope boundaries remain explicit and deliberate. Version 1.0 supports Raydium V4 with WSOL-quote pools on Solana only; additional AMM sources (Raydium CLMM, Orca, PumpSwap, Meteora) are sequenced behind the Phase 15 roadmap, and locker introspection beyond UNCX is tracked as future work. Mainnet activation additionally requires the external audit and bug-bounty program and the multisig custody transition described in Section 14.4. Meanwhile, the EVM-side observatory (Section 4.4) continues to operate as the protocol’s public monitoring surface for Ethereum and Base.

13. Legal and Regulatory Considerations

13.1 The Salvage Permission Spectrum

Ghost pool salvage protocols can be situated on a spectrum defined by who is permitted to initiate salvage and how recovered value is distributed. *Permissioned self-salvage*: only the LP token holder can salvage their own position. Legally simplest but operationally weak — it requires every retail LP to take action and fails the coordination challenge. *Restitution-preserving permissionless salvage*: any party can trigger a salvage, but original LP holders are guaranteed proportional restitution from the recovered value; the salvor earns a finder’s fee for performing on-chain work, and the protocol earns a bounded operational fee. This is GraveYield’s design, and it is the design with maritime salvage law precedent. *Autonomous harvesting*: any party can trigger a salvage, with no restitution to original LP holders; full ownership of recovered value transfers to the salvor and the protocol. Most economically aggressive, legally most exposed, and ethically indefensible.

13.2 Maritime Salvage Law as Legal Defense

The restitution-preserving permissionless salvage model addresses the central legal question raised by permissionless salvage mechanisms through the maritime salvage law analogy. Maritime salvage law establishes three foundational principles directly applicable to the ghost pool context. First, the salvor acts voluntarily without pre-existing duty to the property owner; the salvor’s entitlement to compensation arises from the act of recovery, not from ownership. Second, the original owner retains residual rights to the property after the salvage award; restitution is not optional — it is structural. Third, the salvage award is proportional to the value recovered and the difficulty of the salvage; the 40% salvor share reflects the on-chain labor of pool evaluation across multiple epochs, route construction, gas payment, and transaction submission.

This framework provides a centuries-old, multi-jurisdiction legal precedent for the legitimacy of third-party recovery of abandoned property. It is not a fully settled framework in the context of blockchain-native assets — no court has yet applied maritime salvage principles to AMM LP token positions. The protocol’s legal documentation is explicit on this point: the maritime-salvage concept is an analytical analogy and potential legal theory, not a representation of settled law, and no representation is made that maritime salvage law necessarily applies to decentralized liquidity pools or that participation creates a legally recognized salvage claim in any jurisdiction. What the analogy provides is the strongest available legal frame — one that courts in common law jurisdictions are already familiar with and disposed to respect — and the on-chain design is engineered to match its shape: a formula-compensated finder, structurally guaranteed restitution, and no discretionary actor.

13.3 Residual Legal Exposure

Notwithstanding the maritime salvage analogy, significant residual regulatory exposure exists under several legal theories [13]: conversion (civil tort); constructive trust and fiduciary duty; US state unclaimed property and escheatment statutes; US federal securities law (the Howey test); EU MiCA; and AML/CFT regimes globally. This exposure is acknowledged in full. The protocol’s legal defense rests on the structural commitments embedded in its on-chain logic: guaranteed LP-holder restitution, bounded protocol fee, and the salvor’s finder’s-fee model. The legal documentation set accompanying this revision (v4.0, October 2026) adds the operational instruments that make these commitments legible to counterparties: terms of service, a privacy policy, a protocol risk disclosure, and the multisig charter. Three positions in that documentation merit note. First, the protocol is not represented as a bank, broker, dealer, exchange, custodian, or financial institution of any kind. Second, the protocol operates no token: no \$GRAVE token, security token, NFT program, points program, presale, or token sale exists, and third-party claims to the contrary are explicitly disclaimed — a materially important consumer-protection posture in a token-saturated niche. Third, permissionless smart-contract access is distinguished from interface-level access: the frontend may impose geographic and sanctions-related restrictions even where direct contract interaction is not similarly restricted, aligning the operational surface with AML/CFT expectations without compromising the permissionlessness of the settlement layer itself.

13.4 Regulatory Framework Analysis — Southeast Asia

Singapore’s MAS framework and the Philippines’ BSP VASP framework both focus on custody and exchange functions. A protocol that facilitates atomic on-chain salvage without taking proprietary positions or holding user assets in custodial wallets appears to fall outside the scope of regulated activities under current guidance. This analysis is based on published guidance through October 2026 and may be modified by future regulatory developments.

14. Risk Factors and Limitations

14.1 Data Limitations

The sampling-and-extrapolation methodology yields point estimates with stratum-level confidence intervals of approximately $\pm 15\%$. LP holder accessibility cannot be directly observed from on-chain data. Off-chain criteria are approximations subject to API limitations. The recovery discount factor δ is an assumption, not an empirical calibration. All figures are point-in-time estimates as of May 2026, retained across revisions for comparability; the observatory (Section 4.4) provides

continuous monitoring but is not blended into these point estimates.

14.2 Citation-Verification Status

14.3 Regulatory Risk

The regulatory landscape for DeFi protocols continues to evolve rapidly. The maritime salvage law analogy, while the strongest available legal defense, has not been tested in court in the context of blockchain-native assets. Changes in applicable law or regulatory interpretation in any jurisdiction could require material modifications to any salvage protocol’s design or restrict access for users in affected jurisdictions.

14.4 Implementation and Security Risk

Deployment status does not eliminate implementation risk; it relocates it. Three boundaries merit explicit statement. First, the external security audit and the Immunefi bug-bounty program activate at the devnet-to-mainnet transition; until mainnet, the assurance base is the fork-harness test estate (real mainnet bytecode, adversarial account submission, invariant and fuzz testing) rather than third-party audit. Second, devnet upgrade authority currently rests with a single deployer keypair by design for testnet agility; mainnet requires the fresh, custody-generated keypairs and the Squads multisig transition specified in the pre-mainnet checklist, and no devnet artifact should be trusted as a proxy for mainnet security posture. Third, v1.0 scope limitations remain binding: WSOL-only base tokens, UNCX-only locker introspection (LP locked in other lockers is invisible to on-chain C5 and mitigated SDK-side), the documented oracle honesty and availability boundary, and first-transaction-wins salvage racing. None of these boundaries is silently assumed away; each is tracked as a named item in the specification’s discrepancy ledger.

15. Conclusion and Future Research

15.1 Summary of Findings

This paper has presented the first systematic empirical analysis of ghost pools — stranded liquidity in defunct AMM pools across major blockchain ecosystems. Our analysis identifies approximately 400,000 ghost pools across Solana, Ethereum, Base, BNB Chain, and other EVM networks, containing an estimated \$2.17 billion in stranded base asset liquidity in our central scenario, with a plausible range of \$860 million to \$5.4 billion.

We have demonstrated that ghost pool formation is a structural and enduring feature of permissionless token market ecosystems, driven by the combination of near-zero token creation costs, speculative retail participation, and the behavioral and technical barriers that prevent LP holders from recovering residual positions upon token failure. The resulting stranded liquidity represents a significant and growing source of cap-

Table 11: Citation-verification status.

Citation	Verification status
Chainalysis 2025 [1]	Industry report; verify specific failure-rate figures before external publication.
CoinLaw 2025 [2]	Pre-publication / working-paper status. Verify current version before external publication.
Cong, Li, & Wang 2023 [5]	Authors and journal verified; specific volume/page citation (JFE 147(2), 552–560) should be re-verified.
Hansen, Morris, & Patel 2025 [12]	Working-paper status (MIT Digital Currency Initiative). Verify before external publication.
Pump.fun graduation metrics	Platform-published; subject to historical variation. Cross-check against current Pump.fun documentation.
GraveYield protocol documents [14–16]	Primary-source project documentation (specification, whitepaper, legal documentation); consistent with the frozen specification and devnet deployment record as of October 2026.
Dune observatory [18]	Live public dashboard; figures evolve continuously. Data-division-maintained; re-verify at citation time.

ital inefficiency, with an annualized opportunity cost of \$173.6 million in our central scenario.

We have further demonstrated that automated liquidity salvage infrastructure — implementing restitution-preserving permissionless salvage, cross-chain discovery, multi-epoch eligibility confirmation, and aggregated atomic execution — is technically feasible, legally defensible under the maritime salvage law analogy, and economically significant at scale. This revision converts that demonstration into evidence: the governing specification is frozen, every criterion input now has an on-chain cryptographic evidence path, the full settlement lifecycle has been proven against mainnet bytecode, the complete architecture is live on Solana devnet, and the phenomenon itself is under continuous public observation through the Dune observatory.

15.2 Policy Implications

Regulators evaluating the DeFi ecosystem should be aware that the ghost pool phenomenon represents a genuine market failure with real economic consequences for retail participants. Policy frameworks that facilitate rather than impede the development of salvage infrastructure would serve the interests of retail DeFi participants. In particular, policy treatment of restitution-preserving permissionless salvage — distinct from autonomous harvesting — is an area where regulatory clarity would be valuable. The maritime salvage law analogy provides a ready-made framework for this distinction.

15.3 Future Research

Five directions follow naturally from this revision. First, longitudinal analysis of ghost pool formation and dissolution rates across market cycles, now instrumentable continuously through the observatory rather than requiring repeated sampling campaigns. Second, behavioral economics analysis of LP decision-making in the context of token failure. Third, legal analysis of restitution-preserving permissionless salvage across jurisdictions, exploring the boundaries of the maritime salvage law analogy. Fourth, cross-protocol comparison of ghost pool salvage mechanisms as they emerge. Fifth, macroeconomic analysis of the aggregate effects

of ghost pool salvage on DeFi ecosystem liquidity, including the redeployment dynamics of salvor-earned compensation.

Final observation. The estimated \$5.4 billion upper-bound inventory of stranded DeFi liquidity is not a static artifact of past market excess. It is a dynamically growing inventory, replenished daily by the continuous creation and failure of new token projects. Ghost pools are not an anomaly to be tolerated — they are a solvable market failure awaiting settlement infrastructure. The economic case for building that infrastructure is among the clearest in the current DeFi landscape. The legal and ethical case depends on design: restitution-preserving permissionless salvage, grounded in maritime salvage law precedent, is defensible. Autonomous harvesting is not.

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Appendix A: Data Sources and Methodology Notes

All on-chain data used in this analysis was collected during February – May 2026 using publicly available RPC endpoints, DEX-specific indexing APIs, and custom data collection scripts. Raw data and the sampling-and-extrapolation pipeline are available upon request from the corresponding author at research@graveyield.xyz.

Exchange rate conversions use 30-day average prices ending May 1, 2026: SOL = \$148.20, ETH = \$2,140.50, BNB = \$598.80 (CoinGecko reference prices). Revision r4 retains these reference prices deliberately: the cross-chain estimates of Sections 5–8 are carried forward unchanged from r3 so that revisions remain comparable and the effect of new methodology (rather than new market prices) is isolated. Continuous monitoring of the EVM population is delegated to the observatory (Section 4.4), whose live reserve enrichment supersedes periodic re-pricing between formal sampling revisions; observatory outputs are not blended into the point estimates of this edition.

Appendix B: Ghost Pool Classification Criteria (Academic Definition)

Table 12 restates the five classification criteria of Definition 3.1 together with the data source and the rationale for each threshold. The criteria are designed to be jointly exhaustive and individually cheap to evaluate: the first two criteria establish market death from on-chain evidence alone, while the remaining three screen out pools whose paired project retains observable signs of life.

Appendix C: Academic Definition vs. Protocol-Level Eligibility

Definition 3.1 in this paper is the academic ghost pool definition — it captures the phenomenon comprehensively, including off-chain signals. The GraveYield protocol applies a more conservative on-chain-only subset of these criteria for actual salvage eligibility, together with structural requirements not present in the academic definition. Table 13 presents the mapping as frozen in protocol specification v1.0.0; the direction of every change is toward greater conservatism.

Appendix D: Observatory Instrumentation and Implementation Status

D.1 The Ghost Pool Observatory

The GraveYield Ghost Pool Observatory operationalizes ghost pool monitoring as a public, continuously refreshed dashboard on Dune Analytics. Its query suite screens Uniswap V2 liquidity pools on Ethereum and Base for swap inactivity, then enriches each candidate with latest reserves and token metadata. Table 14 situates its screening threshold relative to the other two definitional layers used in this paper.

The 180-day screen makes the observatory strictly more conservative than the production criterion: any pool it flags has satisfied a no-swap window twice as long as the protocol requires, before any on-chain certification is even attempted. Because the dashboard publishes its underlying queries, third parties can audit the monitored population directly, fork the screens for other DEXes, or diff successive snapshots to estimate formation and dissolution rates — the longitudinal research agenda of Section 15.

D.2 Devnet Deployment Record (9 October 2026)

D.3 Build-Phase Completion Status

Document Changelog

WP-2026-001r4 — October 2026. Integrated the GraveYield Ghost Pool Observatory (Dune) as continuous empirical instrumentation: new Section 4.4

Table 12: Ghost pool classification criteria — academic definition (Definition 3.1).

Criterion	Threshold	Data source	Rationale
Trade inactivity	No swaps in 30+ days	On-chain transaction history	Primary signal of market death.
Token price floor	Below \$0.000001	On-chain reserve ratio + base price	Eliminates micro-cap-but-active tokens.
Developer inactivity	No dev wallet txns in 60+ days	On-chain wallet history	Confirms project abandonment.
Market cap floor	Below \$1,000	Circulating supply $\times$ price	Safety threshold for legitimate micro-caps.
Social inactivity	No social posts in 90+ days	Twitter/X API	Confirms community dissolution.

Table 13: Academic definition vs. GraveYield protocol v1.0.0 (production) eligibility.

Aspect	Academic definition (this paper)	GraveYield protocol v1.0.0 (production)
Trade inactivity	≥ 30 days no swap	≥ 90 days no swap (more conservative), evidenced by oracle-signed attestation
Price signal	Absolute floor $< \$0.000001$	$\geq 99\%$ decline from oracle-attested launch price
Developer wallet activity	≥ 60 days no outbound tx	Not used (off-chain dependency)
Market cap floor	$< \$1,000$	Replaced with minimum residual TVL ≥ 0.5 SOL (pool must hold enough value to justify settlement)
Social media inactivity	≥ 90 days no posts	Not used (off-chain dependency)
LP supply not burned	Not part of academic definition	Required: LP supply $> 1,000$ raw tokens (excludes fully burned pools)
Locked LP	Not part of academic definition	Required: zero locked LP, verified through locker adapter (excludes locked pools that cannot be withdrawn)
Multi-epoch confirmation	Not part of academic definition	≥ 2 consecutive Solana epochs, two-phase certification with criteria-bitmap equality (manipulation defense and LP-holder protection)

Table 14: Inactivity thresholds across the three definitional layers.

Layer	Threshold	Role
Academic definition	≥ 30 days	Classification framework (Section 3)
Protocol production	≥ 90 days	Salvage eligibility, on-chain verified (C1)
Observatory screen	≥ 180 days	Conservative public monitoring (Ethereum, Base)

describes the observatory’s scope (Uniswap V2 on Ethereum and Base, ≥ 180 -day no-swap screen, live reserve and token metadata enrichment) and its methodological relationship to the sampling estimates; observatory rows added to the data-source table (Table 1); new Appendix D.1 situates the 180-day screen against the academic (30-day) and protocol (90-day) thresholds. Deeply integrated the frozen protocol specification v1.0.0 (rev. 1.10.0) into Sections 11–12: the six-criteria table with on-chain evidence sources (Table 9); the two-phase certification lifecycle with criteria-bitmap equality and certificate time-to-live se-

mantics; oracle-attested evidence design (112-byte and 168-byte Ed25519 attestations); settlement mechanics including in-place LP burn, slippage ceiling, conservation invariant, dust recovery, and Merkle-based restitution claims; and Charter-level governance invariants. New Section 12.4 documents implementation status through the 9 October 2026 devnet deployment of GraveScanner and GraveVault with configurations initialized at specification defaults (program IDs in Appendix D.2); new Section 14.4 states implementation and security risk explicitly. Section 13 updated with the legal documentation set v4.0 (terms of service, privacy policy, risk disclosure, multisig charter; no-protocol-token position; interface-level restriction posture). Contribution list (Section 1.1) and future-research agenda (Section 15.3) extended accordingly. May 2026 sampling estimates retained unchanged for cross-revision comparability (Appendix A). Typeset for the first time as a two-column academic working paper; references converted to numeric citations with the citation-verification table retained (Table 11).

Table 15: Devnet deployment record, 9 October 2026. Both programs and configurations were deployed and verified with readback; the emergency-control drill passed on both programs.

Component	Value
Cluster	Solana devnet (api.devnet.solana.com)
GraveScanner program ID	5JiCVxES6RYcrFGnFkqKyDmr7fc3EkYaSCbfgJq7zvNF
GraveVault program ID	HUyoG5vUmYZJDjdBCxRLLAfm98vEXh63WL3pLARox3v6
Scanner ProtocolConfig PDA	Initialized at specification defaults
Vault ProtocolConfig PDA	Initialized at specification defaults
Emergency-control drill	Authority pause → intruder rejection (6000 / 7000) → authority un-pause; readback-verified
Upgrade authority (devnet)	Deployer keypair (single key; mainnet requires custody-generated multisig keys)
Toolchain	Solana CLI 3.0.10 · Anchor 0.32.1 · Rust 1.91.1 · Node 24 LTS

WP-2026-001r3 — May 2026. Aligned with GraveYield Whitepaper v4.0. “Rescue” terminology replaced with “salvage” throughout; “rescuer” replaced with “salvor”; “dead pool” replaced with “derelict pool” in protocol-specific contexts (academic term “ghost pool” retained). Maritime salvage law added as primary legal analogy in Sections 2.3 and 13.2; International Convention on Salvage (1989) added to References. Multi-epoch confirmation added to the technical feasibility analysis and Appendix C. GraveYield protocol description updated: two-phase multi-epoch evaluation flow described; “salvor” and “salvage operation” used throughout. Appendix C updated with multi-epoch confirmation row.

WP-2026-001r2 — April 2026. Aligned with Whitepaper v3.0. Section 11.2 rewritten with 40 / 40 / 20 flat split. Section 13 rewritten with restitution-preserving framing. Upper-bound figure unified at \$5.4B. Citation-verification status table added. Appendix C added.

WP-2026-001 — April 2026. Initial release.

Table 16: Build-phase completion status at revision r4 (October 2026).

Phase	Scope	Deliverable	Status
0	Specification	Frozen <code>PROTOCOL_SPEC</code> v1.0.0 (rev. 1.10.0): criteria, evidence, enforcement matrix, discrepancy ledger	Complete
1.1–1.4	Eligibility evidence	UNCX locker adapter (mainnet-verified 125 locks / 74 pools); C1/C2 Ed25519 oracle attestations; cert lifecycle reissuance	Complete
2.1	Withdraw CPI	Raydium V4 withdraw proven against deployed mainnet bytecode; LP burned in place	Complete
3	Conversion	Jupiter v6 pipeline proven end-to-end, both orientations; slippage ceiling and route-account vetting live	Complete
4	Settlement economics	Dust policy (<code>sweep_dust</code>) and 40/40/20 conservation invariant proven under adversarial configurations	Complete
5.1–5.3	Claims machinery	Deterministic LP-holder snapshotter, Merkle builder, sealed artifacts; claim path proven on real snapshots	Complete
6	Lifecycle integration	First complete lifecycle fork test: candidate → certify → salvage → snapshot → claims	Complete
7	Security hardening	Fuzz, invariant, and adversarial account testing across both programs and the fork harness	Complete
8	Salvor SDK	TypeScript SDK: eight operations, transaction builders, account decoders, PDA derivation	Complete
9	Off-chain indexer	Nine-stage GraveScanner v2 discovery pipeline (Raydium V4), 29 tests, discovery-only default	Complete
11	Devnet launch kit	Real program IDs, identity-gated deployment tooling, rehearsed emergency-control drill	Complete
12–15	Mainnet and expansion	Audit + bug bounty, multisig custody transition, salvor operations, additional AMM sources and lockers	Open